

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12399783
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Zhou; Pin et al.

Partial database restoration

Abstract

Described herein is a system that restores a database by processing a portion of the database. The system restores the database to a previous state at a particular time by reverting data entries that have changed since the time to their initial values before the change. Data entries that have changed after the restore time are identified. For the data entries that have changed after the restore time, their initial values before the change are determined from various sources. The system determines a database version that is created most recently before the restore time. The system additionally identifies changes to the database between the restore time and when the database version is created. The initial values can be determined from either the database version or the changes made to the database between the restore time and when the database version is created.

Inventors:	Zhou; Pin (San Jose, CA), Sarkar; Prasenjit (Los Gatos, CA)
Applicant:	Rubrik, Inc. (Palo Alto, CA)
Family ID:	1000008780912
Assignee:	Rubrik, Inc. (Palo Alto, CA)
Appl. No.:	17/849939
Filed:	June 27, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220327026 A1	Oct. 13, 2022

Related U.S. Application Data

continuation parent-doc US 16151263 20181003 US 11416344 child-doc US 17849939
us-provisional-application US 62567499 20171003

Publication Classification

Int. Cl.: G06F11/00 (20060101); G06F11/14 (20060101); G06F16/21 (20190101)

U.S. Cl.:

CPC G06F11/1451 (20130101); G06F11/1461 (20130101); G06F11/1469 (20130101); G06F16/219 (20190101); G06F2201/80 (20130101)

Field of Classification Search

CPC: G06F (11/1451); G06F (11/1461); G06F (11/1469); G06F (2201/80)

USPC: 714/1-57

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6324548	12/2000	Sorenson	N/A	N/A
6701345	12/2003	Carley	709/205	G06F 16/252
7620785	12/2008	Coulter et al.	N/A	N/A
8244685	12/2011	Fiske	707/684	G06F 11/1469
8478726	12/2012	Habermann et al.	N/A	N/A
9558072	12/2016	Mam	N/A	N/A
9563655	12/2016	Chen et al.	N/A	N/A
10049020	12/2017	French et al.	N/A	N/A
10713124	12/2019	Luo et al.	N/A	N/A
11216344	12/2021	Kochar et al.	N/A	N/A
11416344	12/2021	Zhou et al.	N/A	N/A
2002/0007363	12/2001	Vaitzblit	714/E11.131	G06F 11/1474
2003/0061537	12/2002	Cha	714/E11.13	G06F 11/2094
2003/0191782	12/2002	Buxton	N/A	G06F 11/1471
2003/0220950	12/2002	Hiraoka	N/A	G06F 11/1464
2004/0181560	12/2003	Romanufa	N/A	G06F 16/2365
2005/0081118	12/2004	Cheston	714/47.1	G06F 11/0748
2005/0120064	12/2004	Sinha et al.	N/A	N/A
2007/0083530	12/2006	Lakshminath	N/A	G06F 9/466
2007/0185922	12/2006	Kapoor et al.	N/A	N/A

2007/0294495	12/2006	Uchida et al.	N/A	N/A
2008/0229142	12/2007	Anand	714/6.12	G06F 11/1438
2008/0270838	12/2007	Dorai et al.	N/A	N/A
2008/0285449	12/2007	Larsson	370/232	H04L 49/254
2009/0228429	12/2008	Diaconu	N/A	G06F 16/1865
2009/0323970	12/2008	Cerruti	380/281	G06F 21/602
2010/0017573	12/2009	Shinozaki et al.	N/A	N/A
2011/0087874	12/2010	Timashev	714/E11.02	G06F 11/1469
2011/0161295	12/2010	Ngo	N/A	N/A
2011/0161327	12/2010	Pawar	707/812	G06F 16/907
2011/0258164	12/2010	Mandagere et al.	N/A	N/A
2012/0110287	12/2011	Han et al.	N/A	N/A
2013/0111266	12/2012	Tosaka	714/28	G06F 11/366
2014/0108367	12/2013	Kim	707/703	G06F 16/2379
2014/0115232	12/2013	Goss et al.	N/A	N/A
2014/0250085	12/2013	Sorenson	707/684	G06F 16/2322
2015/0112591	12/2014	Kim	701/532	G01C 21/32
2015/0261623	12/2014	Ho et al.	N/A	N/A
2015/0370644	12/2014	Graefe	707/685	G06F 11/1466
2016/0110268	12/2015	Sekiguchi et al.	N/A	N/A
2016/0117228	12/2015	Farlee et al.	N/A	N/A
2016/0306709	12/2015	Shaull	N/A	G06F 11/1458
2017/0075764	12/2016	Banister	N/A	N/A
2017/0212902	12/2016	Graefe	N/A	G06F 16/113
2017/0230177	12/2016	Barbas et al.	N/A	N/A
2018/0032411	12/2017	Thakur	N/A	G06F 11/1469
2018/0137016	12/2017	Brown et al.	N/A	N/A
2018/0373604	12/2017	Martin et al.	N/A	N/A
2019/0073372	12/2018	Venkatesan et al.	N/A	N/A
2019/0102257	12/2018	Zhou et al.	N/A	N/A
2019/0155699	12/2018	Luo et al.	N/A	N/A
2021/0117293	12/2020	Luo et al.	N/A	N/A
2021/0224166	12/2020	Luo et al.	N/A	N/A
2021/0357297	12/2020	Kochar et al.	N/A	N/A

2022/0058093	12/2021	Kochar et al.	N/A	N/A
2022/0114064	12/2021	Gupta	N/A	G06F 16/219

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
18864993.3	12/2017	EP	N/A
PCT/US2018/054256	12/2017	WO	N/A
WO-2019070915	12/2018	WO	N/A

OTHER PUBLICATIONS

ISA/US, International App. No. PCT/US2018/054256, International Search Report and Written Opinion dated Dec. 4, 2018, 7 pages. cited by applicant

EP Application No. 18864993.3, Extended European Search Report dated Oct. 23, 2020, 8 pages. cited by applicant

Osuna, et al. "Introduction to SnapManager for Virtual infrastructure," SnapManager 2.0 for Virtual Infrastructure Best Practices, IBM Corp., 2010, 6 pages. cited by applicant

Rubrik, "Rubrik Announces Andes 5.0 Release, Enabling Global Enterprises to Protect, Automate, and Govern Mission-Critical Applications and Data Across Hybrid Clouds", www.rubrik.com, 2020, 7 pages. cited by applicant

Rubrik, "Rubrik Andes 5.0: Protect, Automate, and Govern Mission-critical Applications Across Hybrid Cloud", www.rubrik.com, Nov. 12, 2018, 2 pages. cited by applicant

Rubrik, "Game-Changing Innovation for Oracle Data Management," www.rubrik.com, Nov. 12, 2018, 4 pages. cited by applicant

U.S. Appl. No. 62/567,499, filed Oct. 3, 2017. cited by applicant

U.S. Appl. No. 16/151,263, filed Oct. 3, 2018, now U.S. Pat. No. 11,416,344 B. cited by applicant

U.S. Appl. No. 15/819,373, filed Nov. 21, 2017, now U.S. Pat. No. 10,713,124 B2. cited by applicant

U.S. Appl. No. 16/890,542, filed Jun. 2, 2020. cited by applicant

U.S. Appl. No. 17/143,081, filed Jan. 6, 2021. cited by applicant

U.S. Appl. No. 15/931,291, filed May 13, 2020, now U.S. Pat. No. 11,216,344 B2. cited by applicant

U.S. Appl. No. 17/517,895, filed Nov. 3, 2021. cited by applicant

Primary Examiner: Butler; Sarai E

Attorney, Agent or Firm: Holland & Hart LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 16/151,263 by Zhou et al., entitled "Partial Database Restoration", filed Oct. 3, 2018; which claims the priority benefit of U.S. Provisional Application 62/567,499 by Zhou et al., entitled "Efficient Database Restoration to a Previous State Using Change Information", filed Oct. 3, 2017, of which the subject matter is incorporated herein by reference in its entirety.

TECHNICAL BACKGROUND

(1) Database backups can be created periodically to protect the information stored in the databases. In the event that a database crashes, corrupts, or is lost, the backups can be used to restore the

database to an earlier state. For instance, if a database crashes at 4:55 pm, the database can be restored by loading the last backup created at 4 pm. This approach does not restore the database to its latest state before the error. Continue with the previous example, the changes between 4 pm and 4:55 pm are lost. Furthermore, the restoration process is a computing-resource intensive task. Transfer of backups typically consumes a lot of network bandwidth because backups include a large amount of data. In addition, loading a backup into a database consumes a lot of processing resources. During the recovery, the database may not even be accessible, which may lead to reduced productivity for functions that rely on such accesses.

SUMMARY

(2) Described herein is a system that restores a database by processing a portion of the database. The system restores the database to a previous state at a particular time by reverting data entries that have changed since the time to their initial values before the change. As such, the restoration can be achieved by processing the database partially which obviates the need to process the entire database. Compared to conventional ways of restoring databases by loading an entire database backup, the approach described herein is more efficient, and consumes less network bandwidth and computing power. This is because the system does not process those data entries that have not changed since the time.

(3) The system determines a restore time at which the database is to be restored to a previous state determined by the restore time. The restore time can be set to a time point right before when the database crashes, corrupts, or gets lost. These events may be caused by logical errors. Data entries that have changed after the restore time are identified. For the data entries that have changed after the restore time, their initial values before the change are determined from various sources. The system determines a database version that is created most recently before the restore time. The system additionally identifies changes to the database between the restore time and when the database version is created. The initial values can be determined from either the database version or the changes made to the database between the restore time and when the database version is created. The database is restored to the previous state by reverting those data entries to their initial values.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention has other advantages and features, which will be more readily apparent from the following detailed description of the invention and the appended claims, when taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a diagram of an example environment in which a database restore system restores a database by processing a portion of the database, according to one embodiment.

(3) FIG. 2 is a flow chart illustrating an example process of restoring a database by processing a portion of the database, according to one embodiment.

(4) FIG. 3 illustrates a timeline of events in the environment, according to one embodiment.

(5) FIG. 4 is a flow chart illustrating an example process of an environment restoring a database to a previous state by processing a portion of the database, according to one embodiment.

(6) FIG. 5 illustrates an example of an environment determining data entries' initial values, according to one embodiment.

(7) FIG. 6 is a high-level block diagram illustrating an example computing system.

DETAILED DESCRIPTION

(8) Described herein is a system that restores a database to its previous state by processing a portion of the database, which reduces the amount of time and resources used to restore the database. To do so, the system leverages a database system's ability to track changes in a database.

The changes are used to determine which data entries have changed during a time interval. To restore a database back to its previous state at a particular time, only the data entries that changed are rolled back to their previous values, as described in more detail below. This process is much more time and computing resource efficient because the number of changed entries is far fewer than the number of all entries.

(9) FIG. 1 is a diagram of an example environment **100** in which a database restore system restores a database by processing a portion of the database, according to one embodiment. The example environment **100** includes a database restore system **102** and a database system **150**. As further described below, the database system **150** hosts databases and the database restore system **102** manages restoration of the databases hosted by the database system **150**. The database restore system **102** and the database system **150** are in communication with each other via a network **160**. The environment **100** further includes a user device **162** that is in communication with the rest of the environment **100** via the network **160**.

(10) The database restore system **102** manages restoration of databases such as those hosted by the database system **150**. For a particular database that is to be restored to a previous state, the database restore system **102** determines a portion of the database that has changed from the previous state and reverts the portion back to its previous state. The database restore system **102** includes a database version module **104**, a database version store **106**, and a restore data determination module **108**.

(11) The database version module **104** creates and maintains database versions of databases (e.g., the database **152**). A database version captures a state of a database at a particular time. That is, the database version includes a copy of data stored in the database at the particular time point. The data stored in the database includes data entries and associated values. A database version can be a full copy or an incremental copy of the data stored in the database at the particular time point. The full copy includes copies of all data stored in the database at the time point. The incremental copy includes copies of data that has changed since the immediate previous database version was created. The database versions are stored in the database version store **106**. For a particular database, the database versions effectively serve as backups of the database. For example, a previous state of the database can be restored by using a full copy of the database. As another example, a previous state of the database can be restored by using an incremental copy and the full copy of the database referenced by the incremental copy.

(12) The restore data determination module **108** determines the portion of the database that has changed and a previous state to which this portion is to be restored. To determine the portion of the database that has changed, the restore data determination module **108** determines which entries have changed from the previous state. The previous state can be determined by a time. This time is also referred hereinafter as the “restore time.” The database is restored from a later state to the previous state. This later state is a state of the database at a later time after the restore time. In some embodiments, this later state is the current state of the database at the current time.

(13) In some embodiments, the restore data determination module **108** determines the data entries that changed from the restore time to the later time based on change information. The change information describes changes made to the database in a time window. The changes made to the data entries include adding a data entry, deleting a data entry, modifying a data entry, or other changes. In some embodiments, the change information describes a particular data entry that changed and the value of the data entry after the change. The change information also includes a timestamp associated with each change. The restore data determination module **108** uses the restore time and the later time to find the data entries that changed from the previous state to the later state. The data entries that changed during the time interval from the restore time through the later time are associated with timestamps within the time interval. If the later time is the current time, the restore data determination module **108** uses the restore time to find the data entries that have changed from the previous state, for example, by comparing the timestamps associated with the

changes to the restore time. Data entries that changed before the later time can be found in a similar fashion. The change information is created by the database system **150** as further described below.

(14) The restore data determination module **108** obtains the change information from the database system **150**. In some embodiments, if a database is to be restored, the restore data determination module **108** obtains the change information from the database system **150**. In some embodiments, the restore data determination module **108** obtains the change information periodically from the database system **150**. The restore data determination module **108** may obtain the change information describing changes in a particular time interval. In some embodiments, the restore data determination module **108** receives change information from the database system **150** if a change is made to the database. The provision of information can be configured in either a push or pull arrangement between the restore data determination module **108** and database system **150**.

(15) The restore data determination module **108** determines the previous state to which the portion is to be restored by determining initial values of the data entries that changed from the previous state. That is, for each data entry that changed from the restore time, the restore data determination module **108** determines a value of the data entry before the change. This value is also referred to herein as the “previous value” or “initial value.” The restore data determination module **108** determines one or more data sources from where the previous values can be obtained. A data source can be a database version or change information.

(16) The restore data determination module **108** identifies a database version as one data source based on the restore time. For a particular database, the database version is the latest database version that is created most recently before the restore time among all database versions created. This database version is also referred to herein as “a restore database version.” The restore database version is not necessarily the latest database version that is created by the database version module **104** for the database. The restore database version can be identified from database versions stored in the database version store **106**. For example, the restore data determination module **108** compares the restore time to time stamps associated with the database versions stored in the database version store **106** to find the restore database version.

(17) The restore data determination module **108** further identifies any portion of the database that changed from the restore database version creation time to the restore time. For example, any data entry that changed from the restore database version creation time to the restore time is identified. The restore data determination module **108** may identify such data entry by determining change information that describes changes made to the database from the restore database version creation time to the restore time. The determination is similar to the determination of data entries that changed from the restore time to the later time as described above.

(18) For a data entry that changed from the restore time, the restore data determination module **108** determines its previous value from one of the one or more identified data sources. The restore data determination module **108** examines the one or more identified data sources in a reverse chronological order. That is, for a particular data entry, the restore data determination module **108** determines whether the data entry matches any data entry that changed from the restore database version creation time to the restore time. If there is a match, the restore data determination module **108** determines that the value included in the change information is the previous value. If there are multiple matches, the restore data determination module **108** determines that the value in the latest change information is the previous value. If there is no match, the restore data determination module **108** determines that the value of the data entry in the restore database version is the previous value. In some embodiments, when examining whether the data entry matches any data entry in the change information, the restore data determination module **108** stops examining the entire change information after identifying a match.

(19) The restore data determination module **108** provides the previous values of the data entries that changed from the restore time to the database system **150**. To restore the database to the previous state, the database system **150** reverts those data entries to their previous values. That

process effectively restores database **152** to the previous state without having to restore the entire database.

(20) The database system **150** includes a database **152** and a change information store **154**. The database system **150** hosts the database **152**. In operation, the database system **150** hosts information in the form of data entries within the database **152**. The database **152** can be hosted by a single device or a distributed database that is hosted by multiple devices. The database system **150** may include one or multiple devices across which the database **152** is hosted. Users and applications may communicate with the database system **150** to access the database **152**.

(21) The database system **150** tracks changes made to the database **152**. In some embodiments, the database system **150** tracks changes since the creation time of the most recent database version. In some embodiments, the database system **150** tracks changes over time. That is, changes made to the database between database versions are tracked. In various embodiments, the change information is in the form of a change log including change records. A change record is generated to record one change such as a database operation (e.g., insert, delete, update) made to one data entry at a particular time. The change information is stored in the change information store **154**.

(22) The user device **162** is a computing device with which users interact. Via the user device **162**, a user can access the database system **150** and the database restore system **102**. For example, a user can provide the restore time or a logical error via the user device **162**. A logical error may impact one or more database entries, for example, by changing the entries to wrong values.

(23) The network **160** facilitates the data information exchange among different components in the environment **100**. The network **160** includes network elements such as switches, routers, wireless/wired communication links, etc. Many different types of networks could be used. In some cases, the network **160** uses standard communications technologies and/or protocols and can include the Internet, local area networks, and other types of private or public networks. The components can also be connected using custom and/or dedicated data communications technologies.

(24) FIG. **2** is a flow chart of an example process **200** of restoring a database by processing a portion of the database, according to one embodiment. Take the environment **100** illustrated in FIG. **1** as an example, the database restore system **102** determines **202** a restore time to restore the database **152**. The restore time may be determined in a variety of ways. The database restore system **102** may determine the restore time, for example, according to a timestamp associated with a data entry (e.g., when the data entry was created or modified). As one example, a user identifies a corrupt data entry and the database restore system **102** determines the restore time as the time right before the corruption event. The system may also receive the restore time from another system such as the database system **150** or a user device **162**. A restore time may be included in a user instruction.

(25) The database restore system **102** determines **204** data entries that changed after the restore time. Based on the restore time, the database restore system **102** obtains change information that describes changes made to the database after the restore time. For example, the database restore system **102** obtains changes information from the change information store **154** by comparing the restore time to timestamps associated with the changes. As such, the database restore system **102** determines changes made to the database **152** after the restore time. If the database **152** is restored from a later time that is not the current time, the changes are made before the later time. Details of the determination are provided above in connection with the restore data determination module **108**.

(26) The database restore system **102** determines **206** initial values of the data entries before the changes. For example, the database restore system **102** determines one or more data sources that include the initial values of the data entries before the changes. A data source can be a restore database version or change information including changes made to the database from the restore database version creation time to the restore time. For each data entry that changed after the restore

time, the database restore system **120** determines whether it changed from the restore database version creation time to the restore time. If it did, its initial value is the value included in latest change within the time interval between the restore database version creation time and the restore time. If it did not, its initial value is the value stored in the restore database version. Details of the determination of the initial values are described above in connection with the restore data determination module **108**.

(27) The database restore system **102** restores **208** the database to a previous state by reverting the data entries to their initial values. The previous state is the state of the database **152** at the restore time. The database restore system **102** may provide a list of the data entries and their initial values to the database system **150**. For each data entries on the list, the database system **150** reverts its value to the initial value. For example, if a data entry is modified after the restore time, the database system **150** reverts its value to the initial value before the restore time. If a data entry is removed after the restore time, the database system **150** inserts it along with its initial value before the removal. If a data entry is added after the restore time, the database system **150** removes it.

(28) FIGS. **3** and **4** are described in connection with each other. FIG. **3** illustrates a timeline **300** of events in the environment **100**. FIG. **4** is a flow chart **400** illustrating an example process of the environment **100** restoring a database (e.g., the database **152**) to a previous state by processing a portion of the database.

(29) The database restore system **102** periodically creates 401 database versions of the database **152**. As illustrated in FIG. **4**, the database restore system **102** creates database versions every four hours: the database versions **311-315** are created at 12:00 am, 4:00 am, 8:00 am, 12:00 pm, and 4:00 pm, respectively. The database versions can be used to restore the database **152** to a state corresponding to a time point that is between 12:00 am (when the database version **311** was created) and 4:00 pm (when the database version **315** was created). The database restore system **102** may create other database versions over time. Assuming that the database version **311** is the first version created by the database restore system **102**, the database version **311** is a full copy of the database **152**. The database versions **312** through **315** are incremental copies of the database **152**. It should be understood that, while the period between database versions is four hours in this example, other period lengths may be used.

(30) The database system **150** further tracks changes made to the database. The database system **150** creates and stores change information describing the changes made to the database over time. In some embodiments, the change information is stored in a change log including change records. In the illustrated example, after 4:00 pm when the most recent database version **315** was created, the database system **150** creates change records **351** through **355** that describe changes made to the database after 4:00 pm. The change records **351** through **355** each record a data entry change made to a data entry at 4:23 pm, 4:49 pm, 5:01 pm, 5:12 pm, and 5:18 pm, respectively. Each change record includes information such as a data entry identifier identifying the data entry that changes as well as the value of the data entry after the change. The changes made to the database from 12:00 am to 4:00 pm are not shown in FIG. **3**.

(31) The current time **362** is some time after 5:18 pm data entry change which is recorded in the change record **355**. The database **152** is to be restored to a state corresponding to the restore time **361**. The restore time **361** is after 5:01 pm data entry change which is recorded in the change record **353** and before 5:12 pm data entry change which is recorded in the change record **354**. Details of the obtaining information are provided above and are omitted herein.

(32) The database restore system **102** detects **403** a logical error in the database **152**. The logical error began with the 5:12 pm data entry change which is recorded by the change record **354**. The database restore system **102** may receive a report of the logical error from the database system **150**. In some embodiments, the logical error may be detected according to a user input. For example, a user identifies a set of change records that are caused by the logical error and provides the identification to the database restore system **102**.

(33) The database restore system **102** determines **404** a restore time. The restore time may be determined according to a user input based on the time of the logical error. For example, a user identifies database entries that are affected by a logical error and determines the restore time as a time prior to the logical error. The user may identify a series of change records recording the data entry changes caused by the logical error and determines the restore time as the time point before the earliest change record caused by the logical error. The user provides the restore time to the database restore system **102**. In the illustrated example, the 5:12 pm data entry change is caused by the logical error and the restore time **361** is set to a time right before the 5:12 pm data entry change thereby to restore the database to a state before the logical error. The user input may be received from the workstation **303**.

(34) The database restore system **102** identifies **405** data entries that changed after the restore time. For example, the database restore system **102** determines change records associated with time stamps after the restore time. The data entries are identified by data entry identifiers included in the change records. These entries are potentially affected by the logical error. Accordingly, these entries are reverted to their initial values before the logical error. In the illustrated example, the database restore system **102** determines that change records **354** and **355** are associated with timestamps after the restore time **361**, and the corresponding data entries changed after the restore time.

(35) The database restore system **102** determines **406** one or more data sources based on the restore time. The one or more data sources include the initial values of the entries that changed after the restore time. A data source can be a data entry change or a database version. The one or more data sources include the restore database version and any change to the database from the restore database version creation time to the restore time. Based on the restore time, the database restore system **102** identifies the database version that is the latest database version created before the restore time as the restore database version. The database restore system **102** identifies a set of change records associated with time stamps that are between the restore database version creation time and the restore time. For example, the database restore system **102** determines that the database version **315** and the 4:23 pm, 4:49 pm, 5:01 pm data entry changes are the data sources from where the initial values of the data entries can be found.

(36) The database restore system **102** determines **407** the initial values of the data entries from the one or more data sources. For a data entry that changed after the restore time, the database restore system **102** determines a data source that includes the initial value and retrieves the initial value from the data source. For example, for a data entry that changed after the restore time, the database restore system **102** looks up its data entry identifier (ID) in the one or more data sources. If there is a match, the database restore system **102** determines that the matching change record with the latest time stamp stores the initial value. If there is no match, the database restore system **102** concludes that the data entry is added after the restore time and does not exist before the restore time. In some embodiments, the database restore system **102** looks up an entry in the change records in a reverse chronological order. If there is a match, the database restore system **102** terminates checking the rest of the data sources. In some embodiments, to retrieve a data entry's value, the database restore system **102** creates a database index to represent the database and looks up a data entry ID in the database index. The database index is a data structure.

(37) The database system **150** reverts **408** the data entries back to their initial values. The database system **150** receives the initial values from the database restore system **102** and updates the database with the initial values. For a data entry that is identified to have changed since the restore time, the database system **150** replaces its current value with the initial value. For example, if a data entry is removed after the restore time, the database system **150** restores the data entry with the initial value in the database. If a data entry is added after the restore time, the database system **150** removes the data entry in the database. If a data entry is modified, the database system **150** restores the data entry with the initial value in the database.

(38) FIG. 5 illustrates an example **500** the environment **100** determining data entries' initial values. At step **1**, the database restore system **102** determines that data entries **502**, **505** changed after the restore time **361**.

(39) At step **2**, for each data entry, the database restore system **102** determines a data source that stores the data entry's initial value. The database restore system **102** looks up the data entry **502** (or **505**) in a reverse chronological order: from the latest data source (i.e., the change record **355**) to the earliest data source (i.e., the restore database version **315**). Take the data entry **502** for example. The database restore system **102** compares the data entry **502** to the data entry **503** in the change record **355** and does not find a match. After that, the database restore system **102** compares the data entry **502** to the data entry **502** in the change record **352** and finds a match. Because there is a match, the database restore system **102** stops looking up the data entry **502** in the change record **351** or in the database version **315**. The database restore system **102** determines that the value **522** stored in the change record **352** is the initial value of the data entry **502** before the restore time **361**. Take the data entry **505** for example. The database restore system **102** looks up the data entry **505** in the change records **351** through **353**, and does not find a match. The database restore system **102** looks up the data entry **505** in the database version **315** and finds a match. The database restore system **102** determines that the value **551** stored in the database version **315** is the initial value of the data entry **505**.

(40) At step **3**, for the data entry **502**, the database restore system **102** retrieves the value **522** from the change record **352**. For the data entry **505**, the database restore system **102** retrieves the value **551** from the database version **315**. The values **522** and **551** are used to restore the data entries **502** and **505**, respectively.

(41) In one example, the database restore system **102** creates an index indexing data entries in the database version **315** when creating the database version **315**. The database restore system **102** can look up a data entry in the index rather than in the database version. The index of the database version **315** is incorporated into an overall index including indexes of previously created database versions **311-314**. The database restore system **102** uses this index to search for the data entry **505**. Additional discussion regarding this index search method for retrieving data entry values may be found in U.S. patent application Ser. No. 15/409,777, which is hereby incorporated by reference in its entirety. Other manners of obtaining the value **551** may also be used.

(42) At step **4**, the database restore system **102** provides the initial values to the database system **150**. The database system **150** restores the data entries **502**, **505** to initial values **532**, **551**, respectively.

(43) Turning now to a discussion of the implementation of the database restore system **102**, FIG. 6 is a high-level block diagram illustrating an example computing system **600** for implementing the entities shown in FIG. 1. The computing system **600** includes at least one processor **602** coupled with a chipset **604** and the memory **606**. The chipset **604** includes a memory controller hub **620** and an input/output (I/O) controller hub **622**. A memory **606** and a graphics adapter **612** are coupled with the memory controller hub **620**, and a display **618** is coupled to the graphics adapter **612**. A storage device **608**, an input device **614**, and network adapter **616** are coupled to the I/O controller hub **622**. Other embodiments of the computing system **600** have different architectures.

(44) The storage device **608** is a non-transitory computer-readable storage medium such as a hard drive, compact disk read-only memory (CD-ROM), digital video disk (DVD), or a solid-state memory device. The memory **606** holds instructions and data used by the processor **602**. The input device **614** is a touch-screen interface, a mouse, track ball, or other type of pointing device, a keyboard **610**, or some combination thereof, and is used to input data into the computing system **600**. In some embodiments, the computing system **600** may be configured to receive input (e.g., commands) from the input device **614** via gestures from the user. The graphics adapter **612** displays images and other information on the display **618**. The network adapter **616** couples the computing system **600** to one or more computer networks.

(45) The computing system **600** is adapted to execute computer program modules for providing functionality described herein. As used herein, the term “module” refers to computer program logic used to provide the specified functionality. Thus, a module can be implemented in hardware, firmware, and/or software. In one embodiment, program modules are stored on the storage device **608**, loaded into the memory **606**, and executed by the processor **602** to cause the computing system **600** to perform the functions described herein.

(46) The types of computing system **600** used by the entities of FIG. **1** can vary depending upon the embodiment and the processing power required by the entity. For example, the database restore system **102** can run in a single computing system **600** or multiple computing systems communicating with each other through a network such as in a server farm. The computing system **600** can omit some of the components described above, such as the graphics adapter **612** or the displays **618**.

(47) Some portions of the above description describe the embodiments in terms of algorithmic processes or operations. These algorithmic descriptions and representations are commonly used by those skilled in the data processing arts to convey the substance of their work effectively to others skilled in the art. These operations, while described functionally, computationally, or logically, are understood to be implemented by computer programs comprising instructions that are executable by a processor or equivalent electrical circuits, microcode, or the like. Furthermore, it has also proven convenient at times, to refer to these arrangements of functional operations as modules, without loss of generality. The described operations and their associated modules may be embodied in software, firmware, hardware, or any combinations thereof.

(48) As used herein any reference to “one embodiment” or “an embodiment” means that a particular element, feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment.

(49) As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

(50) In addition, use of the “a” or “an” are employed to describe elements and components of the embodiments herein. This is done merely for convenience and to give a general sense of the disclosure. This description should be read to include one or at least one and the singular also includes the plural unless it is obvious that it is meant otherwise.

(51) The descriptions and figures included herein depict specific implementations of the claimed invention(s). For the purpose of teaching inventive principles, some conventional aspects have been simplified or omitted. In addition, some variations from these implementations may be appreciated that fall within the scope of the invention. It may also be appreciated that the features described above can be combined in various ways to form multiple implementations. As a result, the invention is not limited to the specific implementations described above, but only by the claims and their equivalents.

Claims

1. A method, comprising: detecting an error in a database supported by a database restore system; determining a restore time for the database based at least in part on detecting the error, the restore time occurring prior to the error; querying a change log stored at the database restore system to

identify one or more data entries in the database that were modified after the restore time, the change log comprising one or more change records corresponding to the one or more data entries, the one or more data entries comprising a subset of total data entries in the database; determining that a data entry associated with a data entry identifier was added to the database after the restore time based at least in part on the one or more change records not comprising the data entry identifier; determining respective previous values of the one or more data entries from a previous copy of the database stored at the database restore system, the previous copy being based at least in part on a full backup copy of the database and corresponding to a time period prior to the restore time; and performing a partial restoration of the database by reverting the one or more data entries to the respective previous values of the one or more data entries, wherein performing the partial restoration of the database comprises removing the data entry from the database.

2. The method of claim 1, further comprising: identifying one or more partial changes made to the database after the restore time, wherein determining the respective previous values of the one or more data entries is based at least in part on identifying the one or more partial changes.

3. The method of claim 1, further comprising: receiving, via a user interface, a user input that indicates a time at which the error occurred within the database, wherein detecting the error is based at least in part on the user input.

4. The method of claim 1, further comprising: receiving, via a user interface, a user input that indicates a second data entry affected by the error, wherein determining the respective previous values of the one or more data entries comprises determining a value of the second data entry prior to the error, wherein the previous copy corresponds to a time prior to the error.

5. The method of claim 1, further comprising: determining that a second data entry was removed from the database after the restore time, wherein restoring the database comprises adding the second data entry to the database.

6. The method of claim 1, further comprising: retrieving an identifier of a second data entry that was modified after the restore time, wherein querying the change log comprises querying the one or more change records in reverse chronological order using the identifier of the second data entry.

7. The method of claim 1, wherein the one or more change records indicate previous values for the one or more data entries, updated values for the one or more data entries, identifiers of the one or more data entries, and times at which the one or more data entries were modified.

8. The method of claim 1, wherein determining the respective previous values of the one or more data entries comprises: retrieving previous values for the one or more data entries from a plurality of data sources comprising the previous copy of the database within the database restore system.

9. The method of claim 1, wherein determining the respective previous values of the one or more data entries comprises: identifying previous values of the one or more data entries by querying one or more indices corresponding to the previous copy of the database.

10. The method of claim 1, wherein the restore time for the database is selected by a user of the database restore system.

11. The method of claim 1, wherein the previous copy of the database comprises a latest version of the database prior to the restore time.

12. The method of claim 11, further comprising: selecting the latest version of the database from two or more versions of the database stored at the database restore system based at least in part on creation times associated with the two or more versions.

13. The method of claim 1, wherein performing the partial restoration comprises: refraining from processing a remainder of data entries in the database other than the one or more data entries.

14. An apparatus, comprising: a processor; and memory coupled with the processor, the memory comprising instructions that are executable by the processor to cause the apparatus to: detect an error in a database supported by a database restore system; determine a restore time for the database based at least in part on detection of the error, the restore time occurring prior to the error; query a change log stored at the database restore system to identify one or more data entries in the

database that were modified after the restore time, the change log comprising one or more change records corresponding to the one or more data entries, the one or more data entries comprising a subset of total data entries in the database; determine that a data entry associated with a data entry identifier was added to the database after the restore time based at least in part on the one or more change records not comprising the data entry identifier; determine respective previous values of the one or more data entries from a previous copy of the database stored at the database restore system, the previous copy being based at least in part on a full backup copy of the database and corresponding to a time period prior to the restore time; and perform a partial restoration of the database by reverting the one or more data entries to the respective previous values of the one or more data entries, wherein performing the partial restoration of the database comprises removing the data entry from the database.

15. The apparatus of claim 14, wherein the instructions are further executable by the processor to cause the apparatus to: identify one or more partial changes made to the database after the restore time, wherein the respective previous values of the one or more data entries is determined using the one or more partial changes.

16. The apparatus of claim 14, wherein the instructions are further executable by the processor to cause the apparatus to: receive, via a user interface, a user input that indicates a time at which the error occurred within the database, wherein the detection of the error is based at least in part on the user input.

17. The apparatus of claim 14, wherein the instructions are further executable by the processor to cause the apparatus to: receive, via a user interface, a user input that indicates a second data entry affected by the error, wherein the instructions to determine the respective previous values of the one or more data entries are executable by the processor to cause the apparatus to determine a value of the second data entry prior to the error, wherein the previous copy corresponds to a time prior to the error.

18. The apparatus of claim 14, wherein the instructions are further executable by the processor to cause the apparatus to: retrieve an identifier of a second data entry that was modified after the restore time, wherein the instructions to query the change log are executable by the processor to cause the apparatus to query the one or more change records in reverse chronological order using the identifier of the second data entry.

19. A non-transitory computer-readable medium storing code for database restoration, the code comprising instructions that are executable by a processor to: detect an error in a database supported by a database restore system; determine a restore time for the database based at least in part on detection of the error, the restore time occurring prior to the error; query a change log stored at the database restore system to identify one or more data entries in the database that were modified after the restore time, the change log comprising one or more change records corresponding to the one or more data entries, the one or more data entries comprising a subset of total data entries in the database; determine that a data entry associated with a data entry identifier was added to the database after the restore time based at least in part on the one or more change records not comprising the data entry identifier; determine respective previous values of the one or more data entries from a previous copy of the database stored at the database restore system, the previous copy being based at least in part on a full backup copy of the database and corresponding to a time period prior to the restore time; and perform a partial restoration of the database by reverting the one or more data entries to the respective previous values of the one or more data entries, wherein performing the partial restoration of the database comprises removing the data entry from the database.
